

Q1 JEE Main 2020 - 8 Jan (Morning)

If $\int \frac{\cos x \, dx}{\sin^3 x (1 + \sin^6 x)^{2/3}} = f(x) (1 + \sin^6 x)^{1/\lambda} + c$ where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to :

- (A) 4
- (B) -2
- (C) 8
- (D) -4

Q2 JEE Main 2020 - 9 Jan (Morning)

Find integration $\int \frac{dx}{(x-3)^{6/7} \cdot (x+4)^{8/7}}$

- (A) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + c$
- (B) $7\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + c$
- (C) $7\left(\frac{x-3}{x+4}\right)^{\frac{6}{7}} + c$
- (D) $7\left(\frac{x+4}{x-3}\right)^{\frac{6}{7}} + c$

Q3 JEE Main 2020 - 9 Jan (Evening)

If $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \log_e |f(\theta)| + C$ where C is a constant of integration, then the ordered pair $(\lambda, f(\theta))$ is equal to:

- (A) $(1, 1 + \tan \theta)$
- (B) $(-1, 1 - \tan \theta)$
- (C) $(-1, 1 + \tan \theta)$
- (D) $(1, 1 - \tan \theta)$

Q4 JEE Main 2020 - 3 Sep (Evening)

If $\int \sin^{-1}\left(\sqrt{\frac{x}{1+x}}\right) dx = A(x) \tan^{-1}(\sqrt{x}) + B(x) + C$ where C is a constant of integration, then the ordered pair $(A(x), B(x))$ can be

- (A) $(x+1, -\sqrt{x})$
- (B) $(x+1, \sqrt{x})$

(C) $(x - 1, -\sqrt{x})$

(D) $(x - 1, \sqrt{x})$

Q5 JEE Main 2020 - 4 Sep (Morning)

The integral $\int \left(\frac{x}{x \sin x + \cos x} \right)^2 dx$ is equal to (where C is a constant of integration) :

(A) $\tan x - \frac{x \sec x}{x \sin x + \cos x} + C$

(B) $\sec x - \frac{x \tan x}{x \sin x + \cos x} + C$

(C) $\tan x + \frac{x \sec x}{x \sin x + \cos x} + C$

(D) $\sec x + \frac{x \tan x}{x \sin x + \cos x} + C$

Q6 JEE Main 2020 - 4 Sep (Morning)

Let $f(x) = \int \frac{\sqrt{x}}{(1+x)^2} dx$ ($x \geq 0$). Then $f(3) - f(1)$ is equal to :

(A) $\frac{\pi}{12} + \frac{1}{2} - \frac{\sqrt{3}}{4}$

(B) $-\frac{\pi}{6} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

(C) $-\frac{\pi}{12} + \frac{1}{2} + \frac{\sqrt{3}}{4}$

(D) $\frac{\pi}{6} + \frac{1}{2} - \frac{\sqrt{3}}{4}$

Q7 JEE Main 2020 - 5 Sep (Morning)

If $\int (e^{2x} + 2e^x - e^{-x} - 1) e^{(e^x + e^{-x})} dx = g(x) e^{(e^x + e^{-x})} + c$, where c is a constant of integration, then $g(0)$ is equal to :

(A) e^2

(B) 1

(C) 2

(D) e

Q8 JEE Main 2020 - 5 Sep (Evening)

If $\int \frac{\cos \theta}{5 + 7 \sin \theta - 2 \cos^2 \theta} d\theta = A \log_e |B(\theta)| + C$ where C is a constant of integration, then $\frac{B(\theta)}{A}$ can be:

(A) $\frac{2 \sin \theta + 1}{5(\sin \theta + 3)}$

(B) $\frac{2 \sin \theta + 1}{\sin \theta + 3}$

- (C) $\frac{5(2 \sin \theta + 1)}{\sin \theta + 3}$ mathongo mathongo mathongo mathongo mathongo
- (D) $\frac{5(\sin \theta + 3)}{2 \sin \theta + 1}$ mathongo mathongo mathongo mathongo mathongo

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Answer Key

Q1 (B)

Q2 (A)

Q3 (C)

Q4 (A)

Q5 (A)

Q6 (A)

Q7 (C)

Q8 (C)

Q1

$$\sin x = t$$

$$\cos x \, dx = dt$$

$$I = \int \frac{dt}{t^3(1+t^6)^{\frac{2}{3}}} = \int \frac{dt}{t^7\left(1+\frac{1}{t^6}\right)^{\frac{2}{3}}}$$

$$\text{Put } 1 + \frac{1}{t^6} = r^3 \Rightarrow \frac{dt}{t^7} = \frac{-1}{2} r^2 dr$$

$$-\frac{1}{2} \int \frac{r^2 dr}{r^2} = -\frac{1}{2} r + c = -\frac{1}{2} \left(\frac{\sin^6 x + 1}{\sin^6 x} \right)^{\frac{1}{3}} + c$$

$$= -\frac{1}{2\sin^2 x} (1 + \sin^6 x)^{\frac{1}{3}} + c$$

$$f(x) = -\frac{1}{2} \operatorname{cosec}^2 x \text{ and } \lambda = 3$$

$$\lambda f\left(\frac{\pi}{3}\right) = -2$$

Q2

$$\int \left(\frac{x-3}{x+4} \right)^{-\frac{6}{7}} \frac{1}{(x+4)^2} dx$$

$$\text{Let } \frac{x-3}{x+4} = t^7,$$

$$\frac{1}{(x+4)^2} dx = 7t^6 dt$$

$$\int t^{-6} t^6 dt = t + c$$

Q3

$$\begin{aligned}
 & \int \frac{\sec^2 \theta}{\frac{1+\tan^2 \theta}{1-\tan^2 \theta} + \frac{2 \tan \theta}{1-\tan^2 \theta}} d\theta \\
 &= \int \frac{\sec^2 \theta (1 - \tan^2 \theta)}{(1 + \tan \theta)^2} d\theta \\
 &= \int \frac{\sec^2 \theta (1 - \tan \theta)}{1 + \tan \theta} d\theta \\
 & \tan \theta = t \Rightarrow \sec^2 \theta d\theta = dt \\
 &= \int \left(\frac{1-t}{1+t} \right) dt = \int \left(-1 + \frac{2}{1+t} \right) dt \\
 &= -t + 2 \log(1+t) + C \\
 &= -\tan \theta + 2 \log(1 + \tan \theta) + C \\
 &\Rightarrow \lambda = -1 \text{ and } f(\theta) = 1 + \tan \theta
 \end{aligned}$$

Q4

$$\begin{aligned}
 1 &= \int \sin^{-1} \left(\frac{\sqrt{x}}{\sqrt{1+x}} \right) dx = \int \tan^{-1}(\sqrt{x}) dx \\
 &= \int \tan^{-1} \sqrt{x} \cdot 1 dx \\
 I &= x \tan^{-1} \sqrt{x} - \int \frac{x}{1+x} \cdot \frac{1}{2\sqrt{x}} dx \\
 &= x \tan^{-1} \sqrt{x} - \frac{1}{2} \int \frac{\sqrt{x}}{1+x} dx \\
 &= x \tan^{-1} \sqrt{x} - \int \frac{t^2}{1+t^2} dt \quad \left[\begin{array}{l} \text{Put } x = t^2 \\ dx = 2t dt \end{array} \right] \\
 &= x \tan^{-1} \sqrt{x} - \int 1 dt + \int \frac{1}{1+t^2} dt \\
 &= x \tan^{-1} \sqrt{x} - \sqrt{x} + \tan^{-1} \sqrt{x} + c \\
 &= (x+1) \tan^{-1} \sqrt{x} - \sqrt{x} + c \\
 &\Rightarrow A(x) = (x+1), B(x) = -\sqrt{x}
 \end{aligned}$$

Q5

$$\begin{aligned}
 & \int \frac{x^2}{(x \sin x + \cos x)^2} dx \\
 & \because \frac{d}{dx} (x \sin x + \cos x) = x \cos x \\
 &= \int \frac{\frac{d}{dx} (x \sin x + \cos x)}{(x \sin x + \cos x)^2} \cdot \left(\frac{x}{\cos x} \right) dx \\
 &= \frac{x}{\cos x} \left[\frac{-1}{x \sin x + \cos x} \right] - \int \frac{x \sin x + \cos x}{\cos^2 x} \cdot \left[\frac{-1}{x \sin x + \cos x} \right] dx \\
 &= \frac{-x \sec x}{x \sin x + \cos x} + \tan x + C
 \end{aligned}$$

Q6

$$\int \frac{\sqrt{x}}{(1+x)^2} dx \quad (x > 0)$$

$$\text{Put } x = \tan^2 \theta \Rightarrow 2x dx = 2 \tan \theta \sec^2 \theta d\theta$$

$$1 = \int \frac{2 \tan^2 \theta \cdot \sec^2 \theta}{\sec^4 \theta} d\theta = \int 2 \sin^2 \theta d\theta = \int (1 - \cos 2\theta) d\theta$$

$$= \theta - \frac{\sin 2\theta}{2} + c$$

$$\Rightarrow f(x) = \theta - \frac{1}{2} \times \frac{2 \tan \theta}{1 + \tan^2 \theta} + c$$

$$f(x) = \theta - \frac{\tan \theta}{1 + \tan^2 \theta} + c = \tan^{-1} \sqrt{x} - \frac{\sqrt{x}}{1+x} + c$$

$$\text{Now } f(3) - f(1) = \tan^{-1}(\sqrt{3}) - \frac{\sqrt{3}}{1+3} - \tan^{-1}(1) + \frac{1}{2}$$

$$= \frac{\pi}{3} - \frac{\sqrt{3}}{4} - \frac{\pi}{4} + \frac{1}{2}$$

$$= \frac{\pi}{12} + \frac{1}{2} - \frac{\sqrt{3}}{4}$$

Q7

$$\int (e^{2x} + 2e^x - e^{-x} - 1) \cdot e^{(e^x + e^{-x})} dx$$

$$\text{Let } e^x = t, dx = \frac{dt}{t}$$

$$= \int \left(t^2 + 2t - \frac{1}{t} - 1 \right) e^{\left(t + \frac{1}{t} \right)} \cdot \frac{dt}{t}$$

$$= \int \left[\frac{(t^2-1)(t+1)}{t^2} + 1 \right] e^{\left(t + \frac{1}{t} \right)} dt$$

$$= \int \underbrace{\left(t + 1 \right) \left(1 - \frac{1}{t^2} \right)}_r e^{\left(t + \frac{1}{t} \right)} dt + \int e^{\left(t + \frac{1}{t} \right)} dt$$

$$= (t+1)e^{\left(t + \frac{1}{t} \right)} - \int e^{\left(t + \frac{1}{t} \right)} dt + \int e^{\left(t + \frac{1}{t} \right)} dt$$

$$= (e^x + 1) \cdot e^{(e^x + e^{-x})} + c$$

$$\text{So, } g(x) = 1 + e^x \text{ and } g(0) = 2$$

Q8

$$\text{Let } \sin \theta = t$$

$$\Rightarrow \int \frac{dt}{5+7t-2+2t^2}$$

$$\Rightarrow \frac{1}{2} \int \frac{dt}{\left(t + \frac{7}{4} \right)^2 - \left(\frac{5}{4} \right)^2} = \frac{1}{5} \ln \left| \frac{t + \frac{1}{2}}{t + 3} \right| + c$$

$$\Rightarrow B(\theta) = \frac{2 \sin \theta + 1}{2(\sin \theta + 3)} \text{ and } A = \frac{1}{5} \ln \left| \frac{2t+1}{t+3} \right| + c$$

$$\Rightarrow \frac{B(\theta)}{A} = \frac{5(2 \sin \theta + 1)}{2(\sin \theta + 3)}$$